

Alisa Hamilton¹, Alexander Tulchinsky², Cliff Kerr³, Eili Klein^{1,2}, Lauren Gardner¹

¹Johns Hopkins University, ²One Health Trust, ³Institute for Disease Modeling
ahamil33@jh.edu

Definition

GeoPops¹ is an open-source tool for generating geographically and demographically realistic synthetic populations for any US Census area from publicly available data with associated **home, school, workplace, and group quarters (GQ) networks**. GeoPops has the same underlying methodology as GREASYPOP-CO² with the following changes:

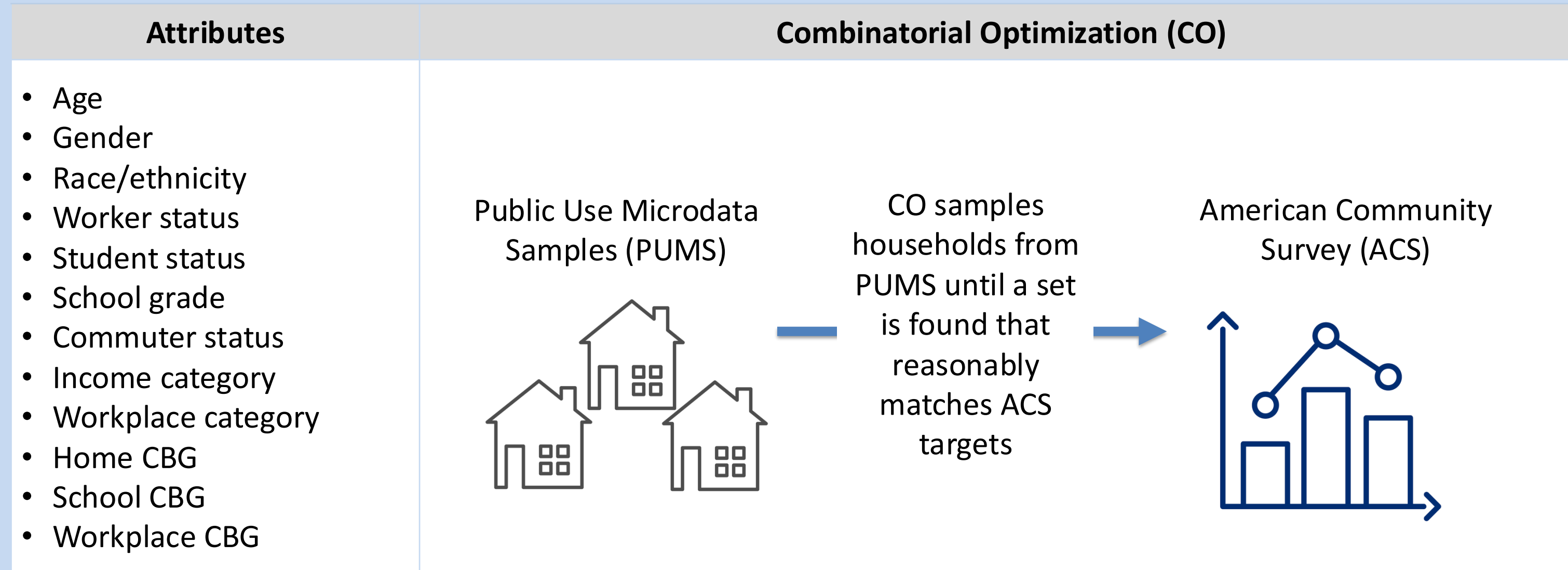
- Wrapped in convenient Python package
- Can be pip installed
- Compatibility with Census data beyond 2019 (still developing)
- Automated data downloading
- Users can adjust all config parameters from the front-end
- Compatibility with the agent-based modeling software Starsim³

github.com/ACCIDDA/GeoPops

pypi.org/project/geopops

Methodology

Step 1: Generate individuals within households



Usage

Code

Example for Howard County, MD

```
import geopops

pars = {'path': 'YOUR_OUTPUT_DIR', # Where you want outputs
        'census_api_key': 'YOUR_KEY', # Census API key
        'main_year': 2019, # main year
        'geos': [24027], # state or county fips of main geo
        'commute_states': [24], # states where people commute
        'use_pums': [24]} # pums state to download

geopops.WriteConfig(**pars) # Set parameters
geopops.DownloadData() # Downloads raw data files
geopops.ProcessData() # Preprocessing for next steps
geopops.RunJulia() # CO, network generation, export to csv
geopops.ForStarsim() # Format for Starsim
```

Step 2: School and Workplace Assignment

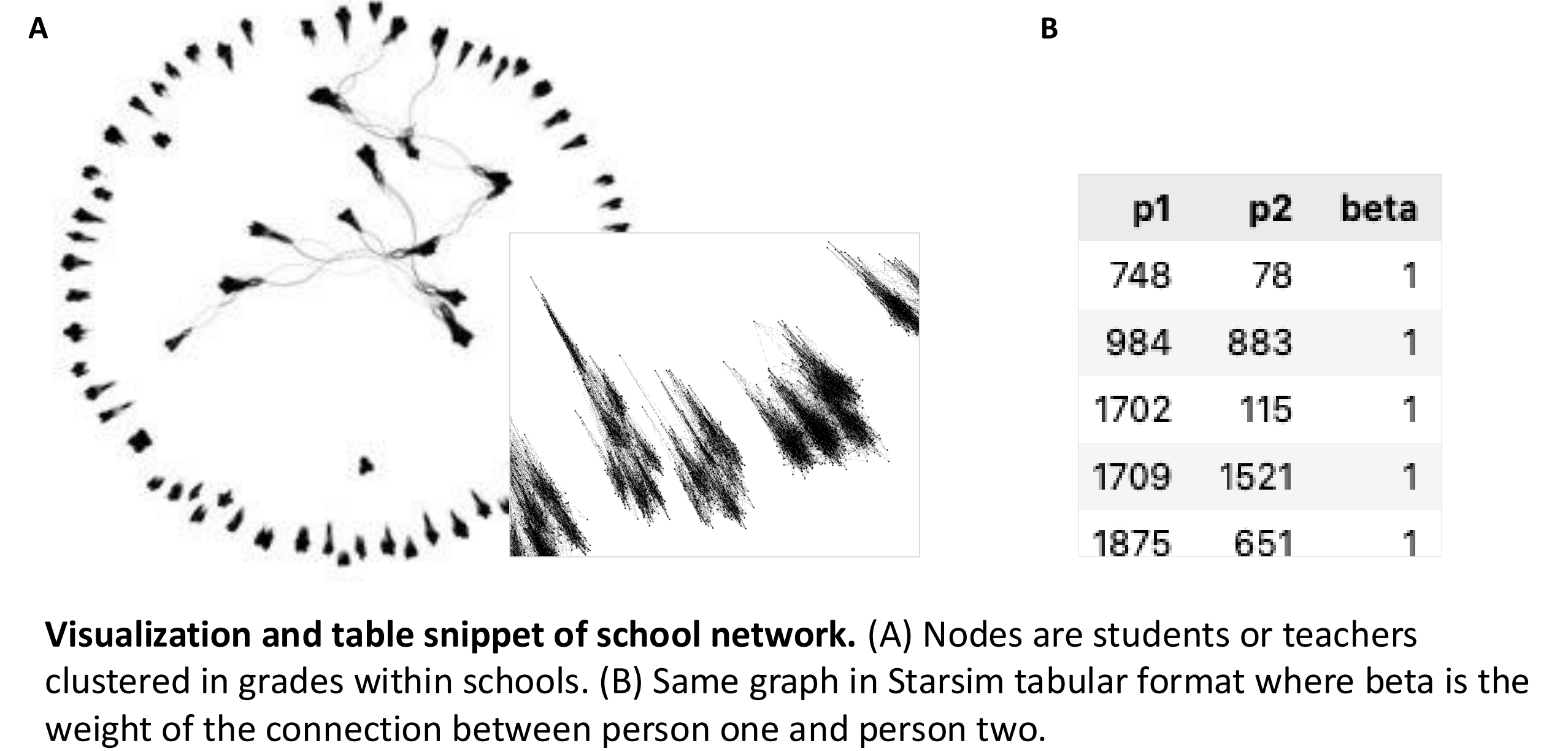
	Methods	Data	Data Sources
Schools	• Students assigned to closest school that offered required grade • Teachers assigned to school by selecting teachers from nearby CBGs	Student and staff counts for each grade level in public schools	National Center for Education Statistics (NCES)
Workplaces	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers	Origin-destination and workplace category matrices	Longitudinal Employer-Household Dynamics (LEHD)

Outputs

- Original files from each data source
- csv of agents and attributes
- csv of homes with locations (CBG level)
- csv of schools with locations and students and teachers by grade
- csv of workplaces with locations and workers by category
- csv of GQ facilities with locations and residents and workers by type (e.g., dormitory vs LTCF)
- csv of outside workers (people who commute into area for work)
- Files in matrix and csv format for each network
- Starsim compatible files for agents and networks

Step 3: Network Generation

Network	Algorithm	Parameters
Household	All agents within a household are fully connected	• N/A
School	Stochastic Block Model within each school	• Assortativity: $\alpha=0.9$ • Mean degree: $K=12$ • Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	• Assortativity: $\alpha=0.9$ • Mean degree: $K=8$ • Blocks are income levels ($<40k$ or $\geq 40k$)
Group quarters	Watts-Strogatz Model within each GQ facility	• Mean degree: $K=12$ • Rewiring probability: $\beta=0.25$



Demonstration (NOT REALISTIC APPLICATION!)

Pass these components into Starsim simulation

GeoPops agents

GeoPops networks

SIR model

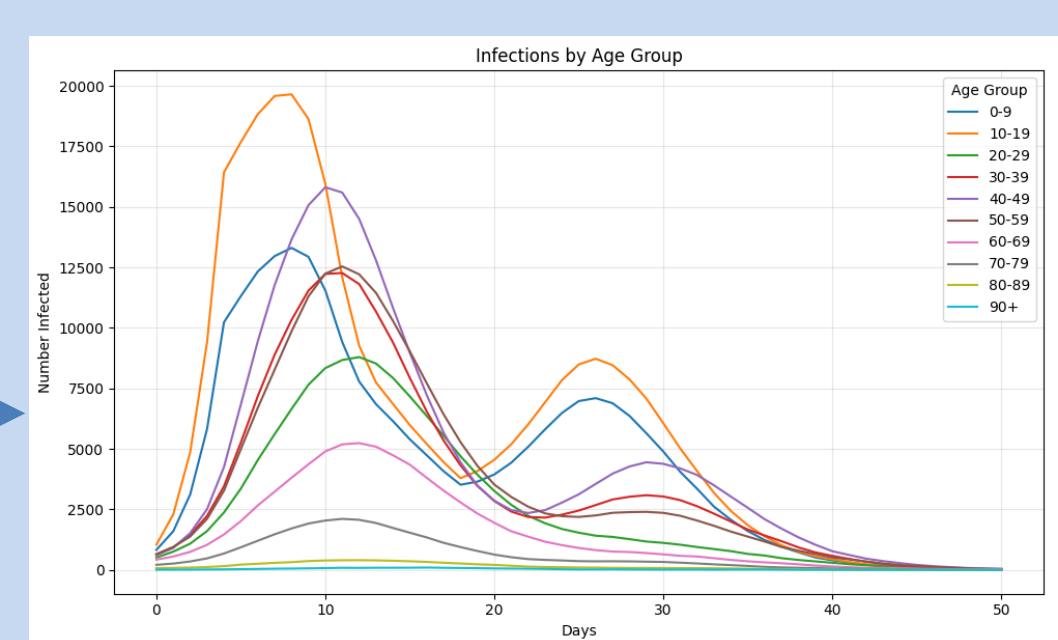
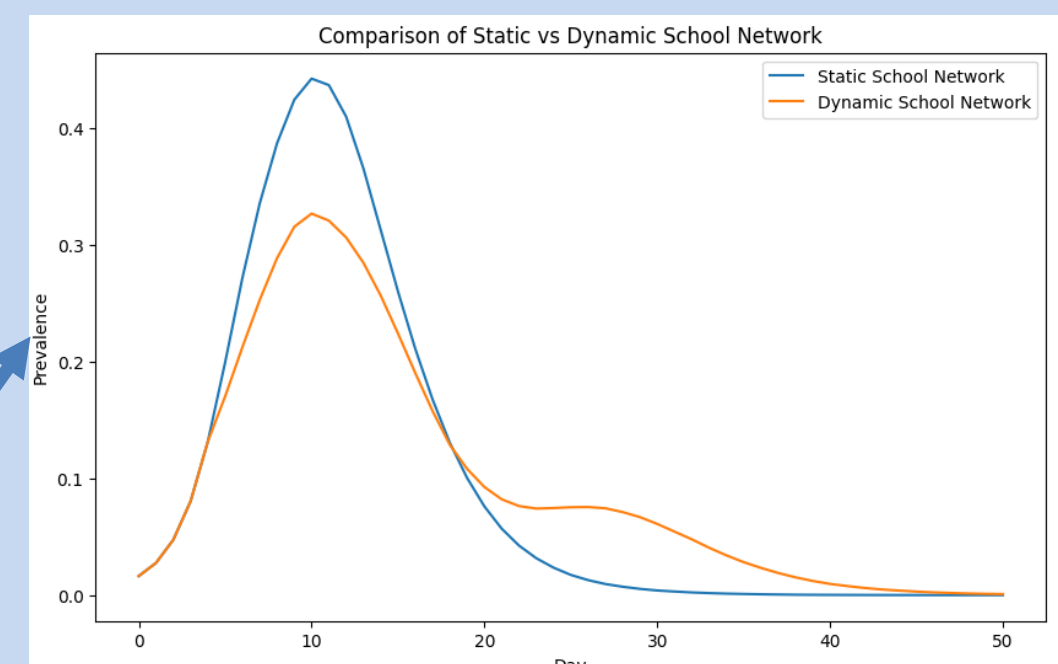
- Respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

Feedback module

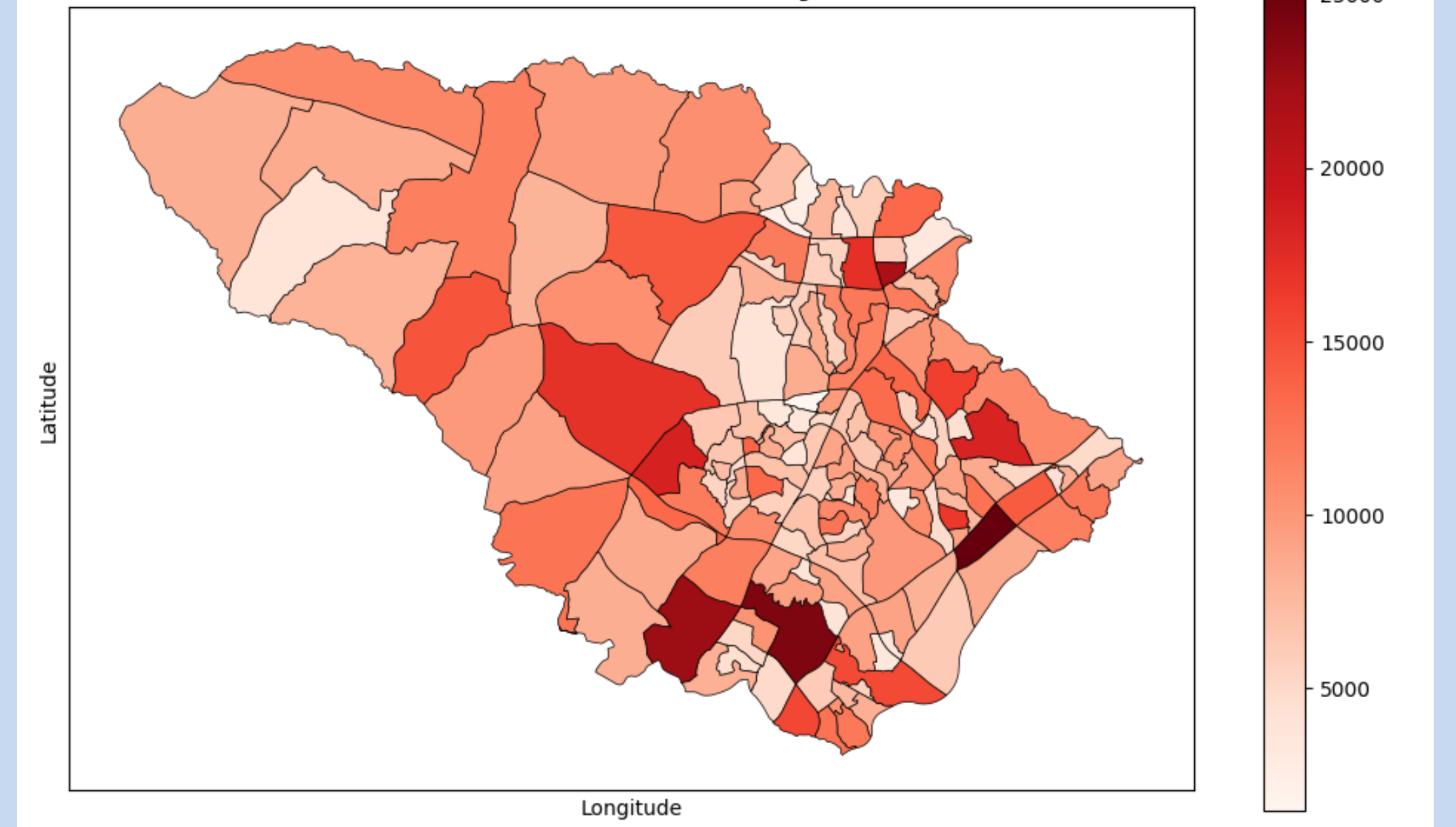
- Removes edges of school network when prevalence > 0.1

Subgroup Tracking

- Infections by age and CBG



Cumulative Infections by CBG



References

Hamilton, Alisa, Sasha Tulchinsky, Gary Lin, Cliff Kerr, Eili Klein, and Lauren Gardner. 2025. GeoPops: Geographically Realistic Synthetic Population Generator. Johns Hopkins University and One Health Trust. <https://github.com/ACCIDDA/GeoPops>.
Tulchinsky, Alexander, Fardad Haghpahanah, Alisa Hamilton, Nodar Kipshidze, and Eili Klein. 2024. "Generating Geographically and Economically Realistic Large-Scale Synthetic Contact Networks: A General Method Using Publicly Available Data." arXiv, ahead of print, June 20. <https://doi.org/10.48550/arXiv.2406.14698>.
Kerr, Cliff, Robyn Stuart, Romesh Abeyuriya, et al. 2024. "Starsim: A Fast, Flexible Toolbox for Agent-Based Modeling of Health and Disease." <https://starsim.org/>.

github.com/ACCIDDA/GeoPops/tutorials/MIDAS

Contact
Alisa Hamilton, PhD(c)
Dept. of Civil and Systems Engineering
Center for Systems Science and Engineering
Johns Hopkins University
ahamil33@jh.edu